Brianna Hinds

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Education

Houston Christian University Houston, TX
Bachelors of Science in Computer Science, Minor in Data Analytics *August 2022 – May 2026*

- **Extracurricular**: Hurdler on the HCU Track Team, President/Founder of SOURCE
- **Relevant Coursework**: Data Analytics, Operating Systems, Data Structures, Database Management Systems

Professional Work Experience

Associate Functional Analyst June 2025 – Present
Sysco *Houston, TX*

- Improved Sysco's data process time by 80% by developing a vector based product matching algorithm to automate item mapping between external vendor data and Sysco's master product dataset.
- Validated the model against manually matched datasets, achieving 100% accuracy.
- Built and evaluated my solution in Amazon SageMaker using Python (sentence-transformers, scikit-learn, pandas) and SQL via DBeaver in data environments.

Data Science Researcher August 2024 – April 2025
Purdue University *West Lafayette, IN, Remote*

- Collaborating with Houston Fire Department on analyzing their station data to improve HFD performance.
- Developing data-driven strategies to optimize resource allocation and response times.
- Applying Python and R, I am implementing data science cleaning, processing, and visualization tools to present the results of the analysis.

Machine Learning Research Assistant May 2024 – August 2025
Houston Christian University *Houston, TX*

- Built a Graph Convolution Networks (GCN) to aid in the prediction of cancers types based on gene expressions. Visualizations such as tSNE were implemented to show the relationship of the data.
- Fine tuned a GCN using Bayesian Optimization to find a classification model architecture of 96% accuracy.
- Applying Python and R, I am implementing data cleaning, processing, and visualization tools to present the results of the analysis.

Projects

Racr: F1 Strategy Platform | *Python, XGBoost, Streamlit Plotly, Pandas, FastF1* January 2025 – Present

- Built an end-to-end race strategy simulation system predicting lap-by-lap and total race time using an XGBoost regression model.
- Engineered race-aware features (tire degradation, stints, pit timing, fuel effects, weather, in/out laps, track characteristics).
- Structured the project using MLOps-aligned practices (feature pipelines, model versioning, cached inference, modular architecture).
- Expanding feature engineering and model validation to improve generalization across tracks and race conditions.
- Achieved high predictive performance ($R^2 = 0.998$) while prioritizing strategy sensitivity and interpretability over pure metric optimization.

Surge | *Python, LangGraph, XGBoost, OpenAI API, Pandas, nmap* June 2025 – May 2026

- Design and implement an intelligent multi-agent cybersecurity framework leveraging large language models (LLMs) to autonomously perform network reconnaissance, vulnerability detection, and executive-level reporting.
- Develop a Vulnerability Agent that integrates the CVE.CIRCL API with a custom XGBoost-based CVSS scoring model, achieving a 99% variance explanation in automated risk classification and prioritization.
- Engineer adaptive data normalization pipelines using Pydantic schemas and LangChain tool binding, enabling seamless interoperability between agents and reducing post-scan formatting errors by 80%.

Technical Skills

Languages: Python, SQL, R, Java, Arduino/C++, HTML, CSS, JavaScript, C
Frameworks/Technologies: Object-Oriented Programming, Machine Learning Models, Neural Networks, PyTorch
Skills: Time Management, Communication, Teamwork, Leadership, Microsoft Excel, Data Cleansing, Git, GitHub